

Enhance Java Applications with FOSS APIs

Java application developers have access to many free and open source APIs that ease their workload. This article, which presumes the reader has some knowledge of Netbeans, Core Java and Java Swing, demonstrates the implementation of user input validation, reporting and SMS integration.



Imagine a world where car manufacturers had to invent the wheel for every car they manufactured, or if Jonas Salk, the inventor of the polio vaccine, had patented his invention or if Dennis Ritchie, the creator of the C programming language, had not created it. What would it be like then? Very different from what it is today, wouldn't it? Thanks to the open source community, we aren't living in such a world.

The Free and Open Source Software (FOSS) community plays an important role in the world of software development. Over the past many years, its efforts and expertise have helped software developers to avoid re-inventing the wheel. Most of the software being developed today uses certain functionalities that have been implemented in some other software. In order to save development time, software vendors break such functionalities into modules and offer them as APIs. They may or may not charge other developers for using these APIs. Those who don't, often make them open source and invite the community to develop them further.

In this article, we explore three such open source APIs for developing Java applications—the SimpleValidation API, MSLSlib API and DynamicReports API. These APIs enable developers to implement user input validation controls, reporting and SMS integration without much overhead.

The following sections cover each of the three APIs in detail. First, they give a brief introduction to the API, followed by its download and implementation details. A sample snippet is included wherever feasible. Each section includes a link to the working example referred to.

The SimpleValidation API

When I was in college, our software engineering professor asked us to remember this simple principle while developing software: 'Always assume that your software will be used by monkeys.' He didn't mean it as an offence, but the message he wanted to convey was that user input validation is mission critical in any software.

An application with relaxed user input validation not only scores badly on the quality front, but is also a security hazard. Therefore, developers tend to spend more time on coding controls for erroneous inputs than on coding actual functionality, irrespective of the programming language or platform used.

Thankfully, for Java application developers, there's good news in the form of the SimpleValidation API, which was developed by Tim Boudreau. It is a Java library developed to ease the coding of user input validation controls. It reduces the time required to code validation controls, thereby allowing developers to enhance the application further, without compromising on project deadlines. Through this library, developers can simply plug the required validation control in most of the swing components. It comes pre-built with some of the common validation controls like the non-empty string, numeric-only, URL, email, length of the input, etc. A more elaborate introduction to and the documentation for the API are both available at <http://goo.gl/MqeQii>.

Download and build

The most recent source code of the API is available at [git://kenai.com/simplevalidation~git](https://github.com/kenai/simplevalidation~git). In order to use this API, you'll need to